

Figure 3: Network graph option in Cooja

Figure 3 depicts the different options by which IoT simulation can be analysed and predictions on various parameters can be made. The network graph tab in the *Collect View* of Cooja enables the researcher to view the dynamic topology and connections of sensor nodes, as shown in Figure 4.

Figure 5 shows the results and parameter based analysis of different nodes in tabular format. It enables the IoT programmer to inspect individual nodes as well as many or all of them in groups, to analyse multiple factors.

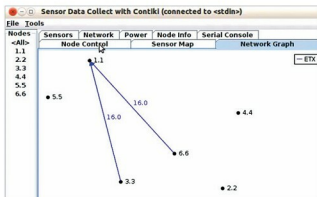


Figure 4: View of the network graph for dynamic topology

Node	Received	Dups	Lost	Hops	RMetric	ETX	Churn	Beacon Interval	Reboots
1.1	0	0	0	0.000	0.000	0.000	0	0	0
2.2	1	0	0	1.000	465.000	16	0	4 min, 22 sec	0
3.3	1	0	0	1.000	627.000	16	0	4 min, 22 sec	0
4.4	1	0	0	2.000	821.000	29	0	4 min, 22 sec	0
5.5	1	0	0	1.000	684.000	16	0	4 min, 22 sec	0
6.6	2	0	0	1.000	640.500	16	0	4 min, 22 sec	0
Avg	1,200	0.000	0.000	1,200	647.500	18	0.000	4 min, 22 sec	0.000

Figure 5: Node information with related parameters

The *Serial Console* tab seen in Figure 6 shows the data transmission log, along with the results related to authentication. In Figure 6, the implementation of the attack and authentication process is depicted, which assists

the developer in examining the effect of customised code on real-time sensors.

Plotting results in the form of graphs

As depicted in Figure 7, there are a vast number of parameters, including LPM, CPU, Radio Listen and Radio Transmit, which need to be looked at during the IoT simulation. The results in the above cited graph are consistent and low-power mode is in the integrity mode. In addition, the Radio Listen parameter is also consistent.

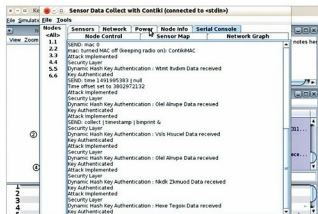


Figure 6: Serial Console to validate the results and key exchange

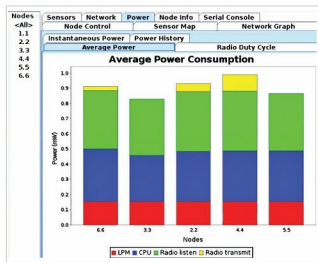


Figure 7: Average power consumption in the motes

Network analysis using Foren6

Foren6 (<http://cetic.github.io/foren6/>) is a powerful tool with a free and open source library. It is used for the analysis of 6LoWPAN networks. It is integrated with the sniffing modules, besides being able to plot the results in assorted formats including text or graphs. Foren6 makes use of integrated sniffers for capturing and analysing the traffic associated with 6LoWPAN in a graphical user interface (GUI). It is powerful enough to record and plot the RPL information with the deep analytics of packets in network transmission.